

## Rebuttal to Reviewer FNYZ

**Additional Baselines.** To further demonstrate the robustness of VaSST, we performed comparisons with PySR (Cranmer, 2023), Operon (Burlacu et al., 2020), and DSR (Petersen et al., 2019) in addition to the competing symbolic regression methods considered in the original submission. Here, we summarize the results corresponding to the Feynman equations  $\Delta U = Gm_1m_2(1/r_2 - 1/r_1)$  and  $P = \kappa A(T_2 - T_1)/d$  as well as a newly added Feynman equation  $f = \beta(1 + \alpha \cos \theta)$ . For all methods, we use  $n = 2000$  (sub-sampled) observations, and reported the out-of-sample RMSEs using the learned symbolic expression on the 10% held-out test set. Different noise regimes have been considered, i.e., from noiseless to noise  $\sigma \in (0.1, 0.2)$ .

VaSST was run with  $K = 3$  trees of depth  $D = 3$ , using the operator set  $\mathcal{O} = \{+, \times, -, /, \exp, \log, \sin, \cos, ^2\}$ . During training, for VaSST we consider reporting the top learned symbolic expression ranked according to the in-sample RMSE. The other hyperparameters are configured the same as in Appendix H.1 of the original submission.

PySR (<https://github.com/MilesCranmer/pysr>) was run with 200 iterations, 10 populations, population size 50, and maximum expression size 20. The operator set was configured to be the same as that of VaSST.

Operon (C++ implementation <https://github.com/heal-research/operon>) was run using its symbolic regressor with allowed symbols  $\{+, -, \times, /, \sin, \cos, \exp, \text{pow}, \text{square}, \text{constant}, \text{variable}\}$ . Specifically, we used `generations = 2000`, `population_size=1000`, `max_length = 20`, and `n_threads = 1`.

DSR was run using the `DeepSymbolicRegressor` (<https://github.com/dso-org/deep-symbolic-optimization>). The operator set used was the same as that for VaSST. The training budget was set to `n_samples = 200000` candidate expressions with `batch_size = 500`. The policy maximum length was set to `max_length = 20`.

**Results.** Across the three Feynman equations considered, VaSST consistently recovers the dominant physical structure, with RMSE generally matching the noise level added to the data. For the trigonometric benchmark  $f = \beta(1 + \alpha \cos \theta)$ , VaSST shows a modest degradation in predictive accuracy, but still achieves strong structural recovery by identifying the core component  $\beta(1 + \alpha \cos \theta)$ . PySR recovers the exact symbolic form in most cases, except for  $f = \beta(1 + \alpha \cos \theta)$ , where it returns a more complex surrogate expression. Operon often attains competitive RMSE, but its learned expressions are typically less interpretable and substantially more complex. DSR successfully recovers the true symbolic expression across all cases, with RMSE close to the corresponding noise level. Detailed results are tabulated below.

Results for learning  $\Delta U = Gm_1m_2(1/r_2 - 1/r_1)$ .

| Method | Learned expression                                                                                                          | RMSE                   |
|--------|-----------------------------------------------------------------------------------------------------------------------------|------------------------|
| VaSST  | $\Delta U = -0.000714 + 0.000914 \frac{G^2 m_1^3 m_2}{r_1^2} - 0.996541 \frac{Gm_1m_2}{r_1} + 1.010681 \frac{Gm_1m_2}{r_2}$ | 0.003656               |
| PySR   | $\Delta U = \frac{Gm_1m_2}{r_2} - \frac{Gm_1m_2}{r_1}$                                                                      | $8.61 \times 10^{-16}$ |
| Operon | $\Delta U = -0.998001 \frac{Gm_1m_2}{r_1} + 1.000232 \frac{Gm_1m_2}{r_2}$                                                   | $1.34 \times 10^{-5}$  |
| DSR    | $\Delta U = \frac{Gm_1m_2}{r_2} - \frac{Gm_1m_2}{r_1}$                                                                      | $8.70 \times 10^{-16}$ |

Table 1: Learned symbolic expressions for Feynman equation  $\Delta U = Gm_1m_2(1/r_2 - 1/r_1)$  in the noiseless setting from VaSST, PySR, Operon, and DSR along with their RMSEs.

| Method | Learned expression                                                                                                                | RMSE     |
|--------|-----------------------------------------------------------------------------------------------------------------------------------|----------|
| VaSST  | $\Delta U = -0.003874 + 0.000609 \frac{G^2 m_1^3 m_2}{r_1^2} - 0.995991 \frac{Gm_1m_2}{r_1} + 1.017123 \frac{Gm_1m_2}{r_2}$       | 0.100194 |
| PySR   | $\Delta U = \frac{Gm_1m_2}{r_2} - \frac{Gm_1m_2}{r_1}$                                                                            | 0.100224 |
| Operon | $\Delta U = -0.998605 \frac{Gm_1m_2}{r_1} + 4.985798 \frac{\frac{Gm_1m_2}{r_1}}{\sqrt{1 + \left(4.921 \frac{r_2}{r_1}\right)^2}}$ | 0.100157 |
| DSR    | $\Delta U = \frac{Gm_1m_2}{r_2} - \frac{Gm_1m_2}{r_1}$                                                                            | 0.09908  |

Table 2: Learned symbolic expressions for Feynman equation  $\Delta U = Gm_1m_2(1/r_2 - 1/r_1)$  under noise  $\sigma = 0.1$  from VaSST, PySR, Operon, and DSR along with their RMSEs.

### Results for learning $P = \frac{\kappa A(T_2 - T_1)}{d}$ .

| Method | Learned expression                                                                                                                                                                                                                                                                                                                                          | RMSE                   |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| VaSST  | $P = 0.014782 - 0.805371 \left[ \frac{AT_1 \kappa}{d} \right] + 0.998520 \left[ \frac{A \kappa T_2}{d} \right]$ $- 0.000254 \left[ \frac{1}{\sqrt{AT_1 \kappa}} - \frac{\frac{d}{\sqrt{A}} - \frac{T_2}{T_1}}{\sqrt{AT_1 \kappa} - \frac{T_2}{T_1}} \right]$                                                                                                | 0.012602               |
| PySR   | $P = \frac{A \kappa (T_2 - T_1)}{d}$                                                                                                                                                                                                                                                                                                                        | $1.59 \times 10^{-16}$ |
| Operon | $P = 4.668 \frac{-2.018665 \left( \frac{T_2}{T_1} \right) (\sqrt{AT_1 \kappa}) + 2.051 (\sqrt{AT_1 \kappa})}{1.904 \sqrt{AT_1 \kappa}} - 9.425 \frac{d}{\sqrt{A}}$ $\sqrt{1 + \left[ \frac{-4.07664 \left( \frac{T_2}{T_1} \right) (\sqrt{AT_1 \kappa})}{\sqrt{1 + \left\{ \sin(-0.172 \frac{T_2}{T_1}) + 0.516 \frac{d}{\sqrt{A}} \right\}^2}} \right]^2}$ | $9.66 \times 10^{-5}$  |
| DSR    | $P = \frac{A \kappa (T_2 - T_1)}{d}$                                                                                                                                                                                                                                                                                                                        | $1.61 \times 10^{-16}$ |

Table 3: Learned symbolic expressions for Feynman equation  $P = \kappa A(T_2 - T_1)/d$  in the noiseless setting from VaSST, PySR, Operon, and DSR along with their RMSEs.

| Method | Learned expression                                                                                                                                                                                                                                                                                                                                       | RMSE     |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| VaSST  | $P = 0.019505 - 0.842846 \left[ \frac{AT_1 \kappa}{d} \right] + 1.005597 \left[ \frac{A \kappa T_2}{d} \right]$ $- 0.000149 \left[ \frac{1}{\sqrt{AT_1 \kappa}} - \frac{\frac{d}{\sqrt{A}} - \frac{T_2}{T_1}}{\sqrt{AT_1 \kappa} - \frac{T_2}{T_1}} \right]$                                                                                             | 0.100325 |
| PySR   | $P = \frac{A \kappa (T_2 - 1.0040724 T_1)}{d}$                                                                                                                                                                                                                                                                                                           | 0.100212 |
| Operon | $P = 0.538 - 0.678 \left[ \frac{\exp(-0.525 \frac{T_2}{T_1}) + 1.888 \sqrt{AT_1 \kappa} - 1.856922 (\sqrt{AT_1 \kappa}) (\frac{T_2}{T_1})}{\sqrt{1 + (1.254 \frac{d}{\sqrt{A}})^2}} \right]$ $+ \sin \left( \sin \left( \exp \left[ \exp \left\{ -1.164401 (\sqrt{AT_1 \kappa}) \left( \frac{T_2}{T_1} \right) \right\} \right] \right) \right) \right]$ | 0.099843 |
| DSR    | $P = \frac{A \kappa (T_2 - T_1)}{d}$                                                                                                                                                                                                                                                                                                                     | 0.09908  |

Table 4: Learned symbolic expressions for Feynman equation  $P = \kappa A(T_2 - T_1)/d$  under noise  $\sigma = 0.1$  from VaSST, PySR, Operon, and DSR along with their RMSEs.

Results for learning  $f = \beta(1 + \cos \theta)$ .

| Method | Learned expression                                                                                                                                                                             | RMSE                    |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| VaSST  | $f = -0.8994 + 0.00125 \cos(2\alpha) + 0.10297\theta$ $+ \underline{1.02384 \beta(1 + \alpha \cos \theta)}$                                                                                    | 0.13497                 |
| PySR   | $f = \beta \left[ 0.009903 \alpha (\theta^2 - 24.691) (7.917 - \theta) + 0.29063 \right] (10.354 - \theta)$                                                                                    | 0.45075                 |
| Operon | $f = 0.214 - 3.021\beta \left[ e^{-0.484\theta} + 0.083\alpha + \frac{e^{-0.500\theta}}{0.080\alpha} \right.$ $\left. - \sin(\sin(0.757\theta)) \{ \sin(0.757\theta) - 0.404\alpha \} \right]$ | 0.66695                 |
| DSR    | $f = \beta(1 + \alpha \cos \theta)$                                                                                                                                                            | $1.848 \times 10^{-15}$ |

Table 5: Learned symbolic expressions for Feynman equation  $f = \beta(1 + \alpha \cos \theta)$  in the noiseless setting from VaSST, PySR, Operon, and DSR along with their RMSEs.

| Method | Learned expression                                                                                                                                                                                                                                                                   | RMSE    |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| VaSST  | $f = -0.92928 + 0.000575 \cos(2\alpha) + 0.10661\theta$ $+ \underline{1.02407 \beta(1 + \alpha \cos \theta)}$                                                                                                                                                                        | 0.16513 |
| PySR   | $f = (-2.48391\theta + 20.96369 + \alpha)$ $\times \beta(\theta - 4.86540)(-0.24211\theta + 1.95075)$                                                                                                                                                                                | 1.49573 |
| Operon | $f = 0.377 - 83.187 \left[ \frac{\cos(-0.981\theta)(-0.368\beta)}{\sqrt{1 + (-1.085\theta)^2}} \right.$ $\left. \times \frac{1}{\sqrt{1 + \left( \frac{\cos(\cos(\cos(-0.999\theta)))}{\sqrt{1 + \exp\{\cos(-0.999\theta) \cos(0.340\alpha)\}^2}} \right)^2}} \right] (0.318\alpha)$ | 0.51157 |
| DSR    | $f = \beta(1 + \alpha \cos \theta)$                                                                                                                                                                                                                                                  | 0.09908 |

Table 6: Learned symbolic expressions for Feynman equation  $f = \beta(1 + \alpha \cos \theta)$  under noise  $\sigma = 0.1$  from VaSST, PySR, Operon, and DSR along with their RMSEs.

| Method        | Learned expression                                                                                                                                                                                                                                     | RMSE    |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <b>VaSST</b>  | $f = -0.95916 - 0.000097 \cos(2\alpha) + 0.11026\theta$ $+ \underline{1.02430 \beta(1 + \alpha \cos \theta)}$                                                                                                                                          | 0.23748 |
| <b>PySR</b>   | $f = \left[ -0.02377 \{ \theta + -0.70132(\theta - 9.13492)^2 \}^2 + 1.22422 \right] \alpha \beta$                                                                                                                                                     | 1.32648 |
| <b>Operon</b> | $f = 3.429 + 0.352 \left[ \left\{ \frac{\cos(0.976\theta)}{\sqrt{1 + (0.653\theta)^2}} \right\} (5.549\beta)(2.578\alpha) \right.$ $\left. + \left\{ \frac{\cos(-0.781\theta)}{\exp(\cos(1.718\theta))} \right\} (-0.437\alpha) - 1.536\alpha \right]$ | 0.46428 |
| <b>DSR</b>    | $f = \beta(1 + \alpha \cos \theta)$                                                                                                                                                                                                                    | 0.1981  |

Table 7: Learned symbolic expressions for Feynman equation  $f = \beta(1 + \alpha \cos \theta)$  under noise  $\sigma = 0.2$  from VaSST, PySR, Operon, and DSR along with their RMSEs.